QUESTION 5: HDFS File Management – Adding Files to HDFS

Aim

To perform file management operations in Hadoop Distributed File System (HDFS) by creating a directory, transferring files from the local file system to HDFS and verifying the uploaded files using Hadoop file system commands.

Procedure

1. First, ensure that Hadoop is properly installed and configured in the system.

2. Open the terminal and start the Hadoop Distributed File System using the following command:

start-dfs.sh

```
3. Check whether the NameNode and DataNode are running properly.
```

jps

4. Create a sample text file in the local file system using the `echo` command.

```
echo "Welcome to Big Data Analytics" > data.txt
```

5. Create a destination directory in HDFS using the `-mkdir` command.

```
hdfs dfs -mkdir -p /user/student/input
```

6. Transfer the local file into the HDFS directory using the `-put` command.

```
hdfs dfs -put data.txt /user/student/input/
```

7. Alternatively, the `-copyFromLocal` command can also be used to transfer files from the local file system to HDFS.

```
hdfs dfs -copyFromLocal data.txt /user/student/input/
```

Note: Use either `-put` or `-copyFromLocal` for the same upload, not both, because the second command may fail if the file already exists.

8. Verify whether the file has been uploaded successfully using the `-ls` command.

```
hdfs dfs -ls /user/student/input
```

9. Display the contents of the uploaded file using the `-cat` command.

```
hdfs dfs -cat /user/student/input/data.txt
```

10. Check the size of the uploaded file using the following command:

```
hdfs dfs -du -h /user/student/input
```

11. To download the file from HDFS back to the local file system, use:

```
hdfs dfs -get /user/student/input/data.txt downloaded.txt
```

12. Finally, verify the downloaded file in the local system.

cat downloaded.txt

Program

HDFS file uploading commands:

Start Hadoop services

start-dfs.sh

Verify Hadoop services

jps

Create a local text file

```
echo "Welcome to Big Data Analytics" > data.txt
```

Create a directory in HDFS

```
hdfs dfs -mkdir -p /user/student/input
```

Upload file into HDFS

```
hdfs dfs -put data.txt /user/student/input/
```

List the uploaded file

```
hdfs dfs -ls /user/student/input
```

Display file contents

```
hdfs dfs -cat /user/student/input/data.txt
```

Check file size

```
hdfs dfs -du -h /user/student/input
```

Download file from HDFS

```
hdfs dfs -get /user/student/input/data.txt downloaded.txt
```

Display downloaded file

cat downloaded.txt

Output of Program

1. Hadoop service verification

```
$ jps
```

```
NameNode
```

DataNode

```
SecondaryNameNode
```

Jps

2. Creating the HDFS directory

```
$ hdfs dfs -mkdir -p /user/student/input
```

Directory created successfully.

3. Uploading the file

```
$ hdfs dfs -put data.txt /user/student/input/
```

File uploaded successfully.

4. Listing the uploaded file

```
$ hdfs dfs -ls /user/student/input
```

```
Found 1 items
```

-rw-r--r-- 1 student supergroup 30

/user/student/input/data.txt

5. Displaying file contents

```
$ hdfs dfs -cat /user/student/input/data.txt
Welcome to Big Data Analytics
```

6. Downloading the file

```
$ hdfs dfs -get /user/student/input/data.txt downloaded.txt
```

File downloaded successfully.

Result

Thus, the file management operation in Hadoop Distributed File System was successfully performed. A local text file was created and uploaded into HDFS using the `-put` command. The uploaded file was verified using the `-ls` and `-cat` commands, and the file was successfully downloaded back to the local file system using the `-get` command.